Hong Kong Institute of Information Technology

Discipline of Information Technology

ITE4116M

Final Year Project

Project Guide

1 Introduction

Final Year Project is a key module in the programme, and is mandatory for all final year students. This module is centered on a single project, chosen by a group of students, which solves a real-life problem using contemporary technologies. The module provides the opportunity for students to link materials from different modules and to reinforce the knowledge gained from the course in previous semesters. Students can further develop their problem-solving abilities and to follow up, in detail, an area of their own interest through the module.

Each student is expected to put in substantial effort in the project. Through active participation in project work, students will be equipped with the abilities in problem identification, the experience in literature searching for knowledge acquisition, the skills in solution formation for the problem based on understanding about the problem, the knowledge for working out the methodology to implement the formed solution, and the hand-on experience in implementing the solution. In addition, good project management is essential to the success of the project. Students are requested to achieve the expected goals (in quality and scope) under the existing constraints (especially in time and resources).

Students should investigate possible solutions to their selected problem and develop a plan for the implementation of an environmental friendly solution. Students should also explore the possibility to develop a bilingual or cross-border deployable system to meet Greater China business needs. Also, students are required to develop their project professionally and ethically and according to the codes of conduct of leading professional bodies of related disciplines.

2 Learning Outcomes

Through active participation in project and completion of the project, a student should be able to:

- research on a reasonably complex business or technical problem and propose a possible solution in the area of information technology
- develop a project development plan, and monitor and report their project progress
- analyze, design, implement, and test a proposed solution
- work independently and collaboratively with others as a member of a project team
- document and verbally present their work in a professional way

3 Project Formulation and Development

Project topics will normally be set so that students can work together as a group to collectively develop a solution. Each project team involves 3-4 students. However, each student will have their own assigned roles and responsibilities and their contributions will be individually assessed. The distinguishable responsibilities and works of each student

should be clearly specified in the project proposal and progress reports. Marks will be awarded to the members according to their contributions to the project.

The approaches to divide the work of developing a large system:

- data interfaces – divide the system into several subsystems. The subsystems communicate with each other by a database. Issue: must agree on a common database.
- programming interfaces – divide the system into several subsystems. The subsystems communicate with each other by calling functions. Issue: must agree on the specifications of functions provided by each subsystem.
- protocol interfaces – divide the system into several subsystems. The subsystems communicate with each other by sending messages. Issue: must agree on the communication protocol.

3.1 Project Selection Procedure

1. The students form a project group.
2. The students think about what they want to achieve to satisfy the objectives of the unit.
3. The students discuss the ideas with a lecturer.
4. The lecturer agrees to supervise the student. The project group must send an email which includes the member names, group & group numbers and the supervisor name to the project coordinator ***and*** the supervisor.
5. The students write and submit a project suggestion to the supervisor.
6. The students write and submit a project proposal to the supervisor.
7. If the proposal is not approved by the supervisor, repeat step 5.
8. The proposal is accepted.

3.2 Milestones

- Sep — 1) Project Group Member List sent to the Project Coordinator *and* Supervisor.
  2) Project Suggestion submitted to Supervisor.
- Sep — Submit Project Proposal
- Nov — Submit Initial Report
- Jan/Feb — Submit Interim Report and prepare Oral Presentation covering project plan, requirements, analysis, design, suggested solution and prototype implementation.

- April/May — Submit Final Report and prepare Oral Presentation & Demonstration covering requirements, analysis, design, solution, implementation, program listing, user guide, and installation guide.

3.3 Industrial or Commercial Based Projects

Projects undertaken in conjunction with industry are encouraged. In such situations, the IVE-based Project Supervisor must be satisfied that the student has adequate supervision and that the assessment process and standard are not compromised.

- Students are encouraged to propose 'real-world' projects.
- **Students are advised not to** promise to deliver any software or documentation to the external organization without the supervisor's consent. Students may not be able to finish your work according to the deadlines set by the external organization.
- At present, IP rights associated with 'in-house' student projects reside with the students, and VTC is automatically sub-licensed to use that IP.
- With external projects, students may generally be required to sign an agreement with the sponsor organization requiring them to relinquish your IP rights. Students should weigh the potential benefits of working on a real industrial project the loss of their IP rights.
- Students may also be asked to sign a Non-Disclosure Agreement form which requires them to protect the confidential information provided by company. Please use the standard Non-Disclosure Agreement Form prepared by the department obtainable from the project coordinator as far as possible.

4 Project Management

The Final Year Project does provide a precious opportunity for the students to practice and strengthen their knowledge and skills in project management acquired in their previous studies.

The project management skills being in concern are:

- Planning
- Problem handling (ability in solving problems independently)
- Progress management (keeping the decided schedule and appropriate modification of the schedule)
- Resources allocation (balanced distribution of effort among different activities)
- Communication among different teams members and parties
- Co-operation of team members

Before actually implementing the solution for the problem, a development schedule should be drafted first, and the students are requested to follow the schedule in a serious manner so as to prevent the students from developing the system in an ad-hoc manner or in rush during the period close to the project submission deadline

4.1 Project Review

The students should review the project regularly so as to:

- adjust the development schedule according to the actual situation (for each of such adjustment, description of the cause / issue behind should be logged and presented in the final report)

- allot further details to develop plan of later development activities (the ground supporting such further details should be logged and presented in the final report)

The students should meet their supervisor for 30 minutes every week.

4.2 Supervisor and Co-marker

Each project is assigned an overall supervisor and a co-marker to assist with assessment. It is the supervisor's role to act as a mentor to the student and to guide the planning of the project solution.

5 Deliverables

5.1 Initial Report

In semester 4, students are required to submit an initial report which should cover the analysis work undertaken, requirement specification, initial design and project plan.

The report may include, but not limit to:

- Background information of the problem
- Expected stakeholders of the project
- Assumed users of the completed system
- Exploration of possible solutions
- Overview of the Proposed solution
- Expected scope and limitations of the proposed solution (specification of the system)
- Required resources (hardware and software) for the system development
- Expected deliverables to be generated
- Project schedule:
  - Milestone / expected deliverables of each development activity
  - Expected completion dates of early development activities
- Task and role of each student
- References

The report must be well-presented.

5.2 Interim Assessment

At the beginning of semester 5, students are required to submit an interim report which includes a written submission, an oral presentation and a demonstration of prototype work undertaken. The interim report should cover analysis work, requirement specification, initial design, prototypes, etc. and will be assessed according to the quality of work done so far.

5.2.1 Interim Report

At the middle of the project development, an interim report should be submitted.

The report may include, but not limit to:

- Requirement analysis:
  - General descriptions of the existing systems, the intended users, and its relationship to other systems
  - General descriptions of all the functions of the system
  - High-level diagrams, e.g. systems architecture, context diagrams, etc.
  - Data dictionary for the entire user level data
- Requirement specification/Detailed Design:
  - Data models and process models being employed in the solution
  - Physical configuration of the system under construction
  - Function architecture of the system under construction (functions to be provided by each module in the system)
  - User interface design
- Problems/difficulties encountered in the due course of system development, and solutions being employed for overcoming the problems / difficulties
- Further details allotted to the initial project plan as described in the initial report
- Deviation of the project progress from the originally proposed schedule, and the causes behind such deviation
- Test plan

Students adopting Object-Oriented methodologies for system analysis and design should provide the following set of UML diagrams and documents in the interim and subsequent reports:

- Requirement analysis:
  - Actor descriptions
  - Use case diagram
  - Initial use case descriptions and base use case descriptions
  - Data dictionary
- Requirement specification/Detailed design:
  - Class diagram (design level)
  - 3-tier sequence diagrams
  - State diagrams for control objects of some use case

The report must be well-presented.

5.2.2 Interim Oral Presentation

At the middle of the project development, students are required to give an oral presentation of their progress.

During the presentation, students should at least show the following aspects concerning the project:

- Project background
- Encountered problems and employed solutions for problem solving
- So far completed work
- Expected upcoming progression
- Prototype demonstration
- Insufficient aspects in the completed work, expected improvements in the upcoming development

In addition, students are expected to handle or answer questions concerning the project.

For the presentation, assessment will be based on the following factors:

- Selected material for the presentation
- Preparation for the presentation
- Presentation skill and question handling skill

5.3 Progress Report and Mid-Semester Demonstration

To ensure your project is on the right track, you are required to submit a progress report and conduct a mid-semester demonstration to your supervisor at the middle of semester 5.

The progress report should describe what have been achieved during the reporting period and contain a list of outstanding tasks and system functions.

5.4 Final Assessment

At the end of semester 5, students are required to submit the final product and a final report that presents and demonstrates the completed project. An oral presentation and a demonstration of the final product is also required.

5.4.1 Final Report & Completed System

At the end of the project development, a final report should be submitted.

The report may include, but not limit to:

- Final design of the system, deviation of the final design from the original design, and the ground for such deviation
- Final version of the test plan
- Test results in each test
- Final version of the project development plan having all details allotted
- Actual progress of the project, deviation of the progress from the schedules as mentioned in previous reports, and causes behind such deviation
- User manual of installation and operation of the completed system
- Critical Evaluation of the Project

The report must be well-presented.

5.4.2 Final Presentation and Demonstration

After the project development period, each project group should have a presentation and demonstration on the completed work.

During the session, the group will have an oral presentation at least on the following aspects:

- Description of the existing problem
- Formed solution for the problem
- Scope and limitation of the solution
- Rooms for further improvement on the completed project / system

The project group should conduct also a demonstration of the completed system, which is used to show the features/functionality of the completed system.

In addition, students are expected to handle or answer questions concerning the project.

For the presentation and demonstration, assessment will be based on the following factors:

- Selected material for the presentation and demonstration
- Preparation for the presentation and demonstration
- Design of the operation flow in the demonstration
- Presentation skill and question handling skill

5.5 Project Management Skills and Logbook

Project management skills are assessed according to the management of project work by the student over the whole project period and are based on project logs. Students' performance will be monitored through regular meetings with their project supervisor. Students will be assessed on their information searching skills, project management skills and effort. They have to keep a log of their activities which can be regularly reviewed.

These regular meetings will also serve as the forum to maintain a record of the contribution of individual students to the group project.

5.5.1 Logbook

Each group participating in the Final Year should submit a logbook, which is composed of a number of bi-weekly log-sheets.

Alternate week, each group should fill in a log-sheet with the following entries:

- items completed in the past one week
- to-do list
- findings in background research information (optional)
- list of collected documents and reference materials (optional)
- innovated ideas (optional)

Rationales behind having such a logbook for each group are as the followings

- provide a repository of project data, which facilitates the authoring of reports
- provide background data for further project plan detailing and regular project review
- ensure even progression of the system development and prevent students from working on the project in rush within the period close to the system submission deadline
- prevent existence of free-riders through recording contributions of each student

6 Assessment

Assessment is staged, in order to provide checkpoints for monitoring progress. There are four assessment components and the weightings are as below:

Assessment Components	Weighting
▪ Initial report	10%
▪ Interim report	30%
▪ Final report	50%
▪ Project management skills	10%

Each student in the project group will be assessed individually on his/her performance in project management skills, oral presentation/demonstration and implementation quality of the subsystem of the final product that the student is responsible for. The initial report will be assessed according to group performance. The written submission of the interim report and the written submission of the final report will be assessed according to group performance and individual contribution. The assessment weighting for collaborated submission is 6:4 in favour of individual contribution (i.e. 60% for individual mark and 40% for group mark). At each stage, a project group has to submit a written statement which indicates the individual contribution of work of each student in the group in that stage. The supervisor and the co-marker will consider these percentages and then determine the weighting given to each student for awarding a staged mark. Mark will be firstly awarded using the same marking scheme for individual projects. The intermediate

mark is then redistributed to give the mark of each student in a group according to the following scheme:

1. Group mark for each student: marks of the deliverable $\times 0.4 \times (1 - p)$
2. Individual mark for a student: marks of the deliverable $\times 0.6 \times n \times \dfrac{w_i}{w_1 + w_2 + \quad + w_n}$

   where n is the number of students in the group

   w_i is the weighting given to student i and $w_1 + w_2 + \quad + w_n = 100$
3. The mark for a student = group mark for the student + individual mark for the student

The following table summarizes the break-down of group performance and individual contribution in this assessment scheme:

Annex 1: Project Proposal Guidelines

Assessment Components	Weighting	Assessment criteria	
		Group Performance	Individual Contribution
Initial Report	10%	10%	0%
Interim Report • Written submission • Prototype implementation • Demo and oral presentation	 15% 10% 5%	6%	 9% 10% 5%
Final Report • Written submission • Final Product • Demo and oral presentation	 15% 30% 5%	6%	 9% 30% 5%
Student Performance, including information search, management skills, effort etc.	10%	0%	10%
Total:	100%	22%	78%

Appendix

Format of the Report

Size and typing: single line or 1.5 lines spacing throughout; A4 white paper; a font size of 12 point for the body text.

Margins: all margins are at least 1 inch, except the left margin, which is 1.5 inches. This is to allow for binding. Spiral binding has to be used.

Page numbering: the pages are numbered consecutively except for the title page. Appendices are numbered separately on each page as A1, A2, B1, B2 etc.

Style of writing: it has become conventional for report writings to refer to oneself in the text as 'the author' rather than 'I' or 'me'. Mis-spelling is irritating - use the spell checker inside the word processor.

Source Materials: extracts and interpretations of other people's work should be identified in the text by either the authors' name(s) and/or a numeric index; the source of the extract or interpretation should be listed under the section headed *References*. This section should list all references in alphabetic order of author's surname. The detail must be given to enable the reader to trace the source; hence it must contain the following elements in sequence:

(a) the author's surname followed by his initials (e.g. SMITH, A B).
(b) the date of publication (e.g. 1985)
(c) the title of the work (in italic)
(d) the place of publication (e.g. London)
(e) the name of publishers (e.g. Wiley)

Examples:

1. LANGEFORS, B. (1973). *Theoretical Analysis of Information System*, Philadelphia: Anerbach.
2. SALTON, G., YANG, C. S. AND YU, C. T. (1975). *A theory of term importance in automatic text analysis*, Journal of the American Society for Information Science **26**, 33-44.

Organisation: The report must be divided into major sections, each with its own heading, beginning on a fresh page. The appendix has its own main heading, and each appendix is appropriately labelled: 'APPENDIX A', 'APPENDIX B'

Tables and diagrams: should be numbered 'Table 1', 'Figure 1' throughout the text.